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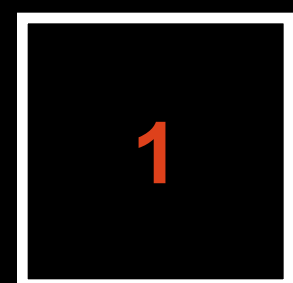
Introduction to DevOps

Endava DevOps Team



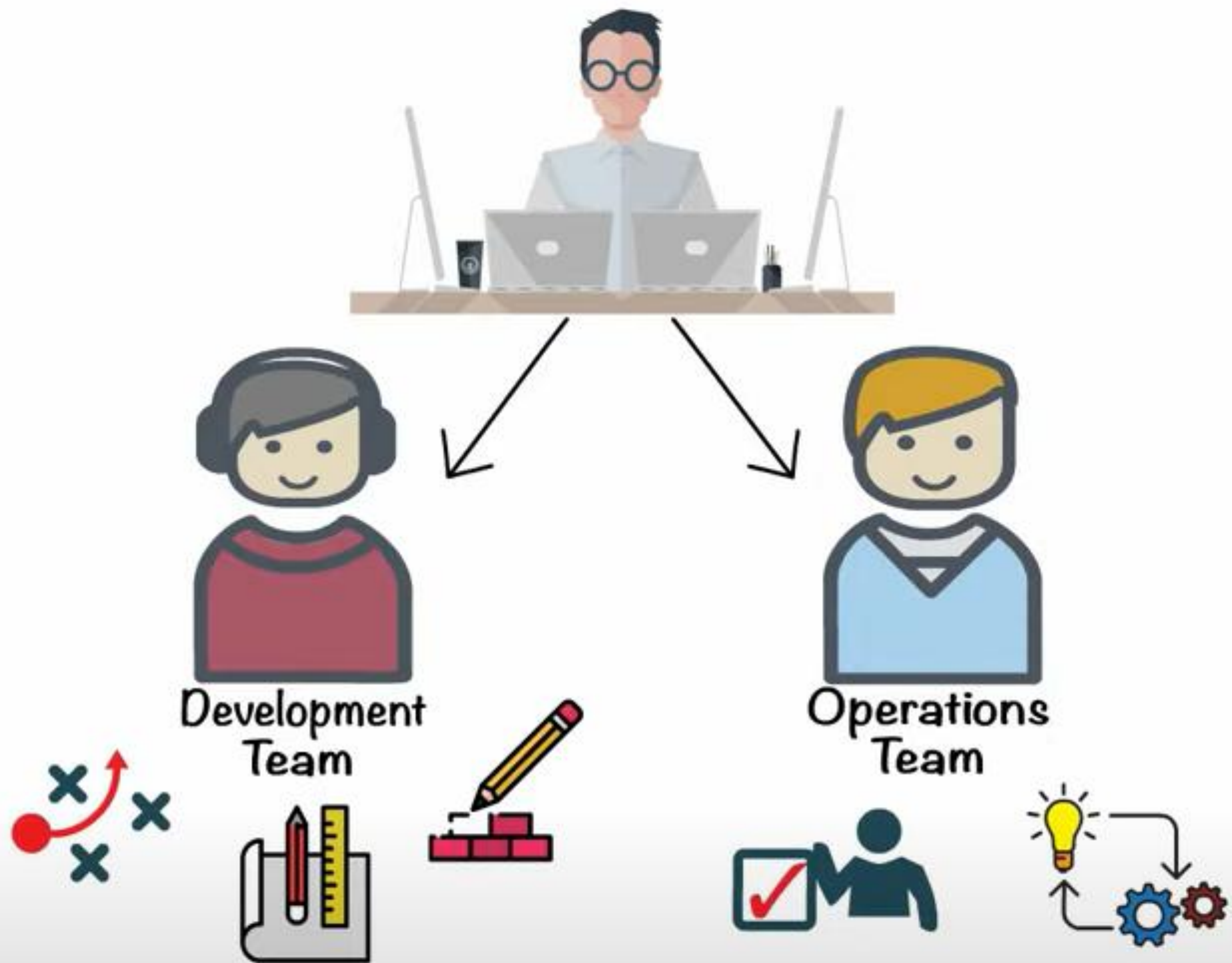
Agenda

1. SOFTWARE DELIVERY PROCESS
 - WATERFALL
 - AGILE OVERVIEW
 - OPERATIONS MANAGEMENT
 - CONFIGURATION MANAGEMENT
 - WHAT IS DEVOPS
2. INTRODUCTION TO CI/CD
 - WHAT IS CI/CD AND DEVOPS FLOW
 - OLD WAY-NEW WAY
 - DEVOPS PRINCIPLES
 - DEVOPS ADVANTAGES
3. PROCESS AND TOOLING
 - LIFECYCLE MANAGEMENT
 - CONFIGURATION MANAGEMENT
 - SOURCE CONTROL MANAGEMENT
 - CI TOOLS (AZURE DEVOPS / GITHUB PIPELINES)
 - BUILD TOOLS
 - STATIC CODE ANALYZE (SONAR)
 - ARTIFACTORY (NEXUS, AZURE DEVOPS ARTIFACT)
 - DEPLOYMENT
 - TESTING TOOLS
 - LOGGING & MONITORING
4. Q&A

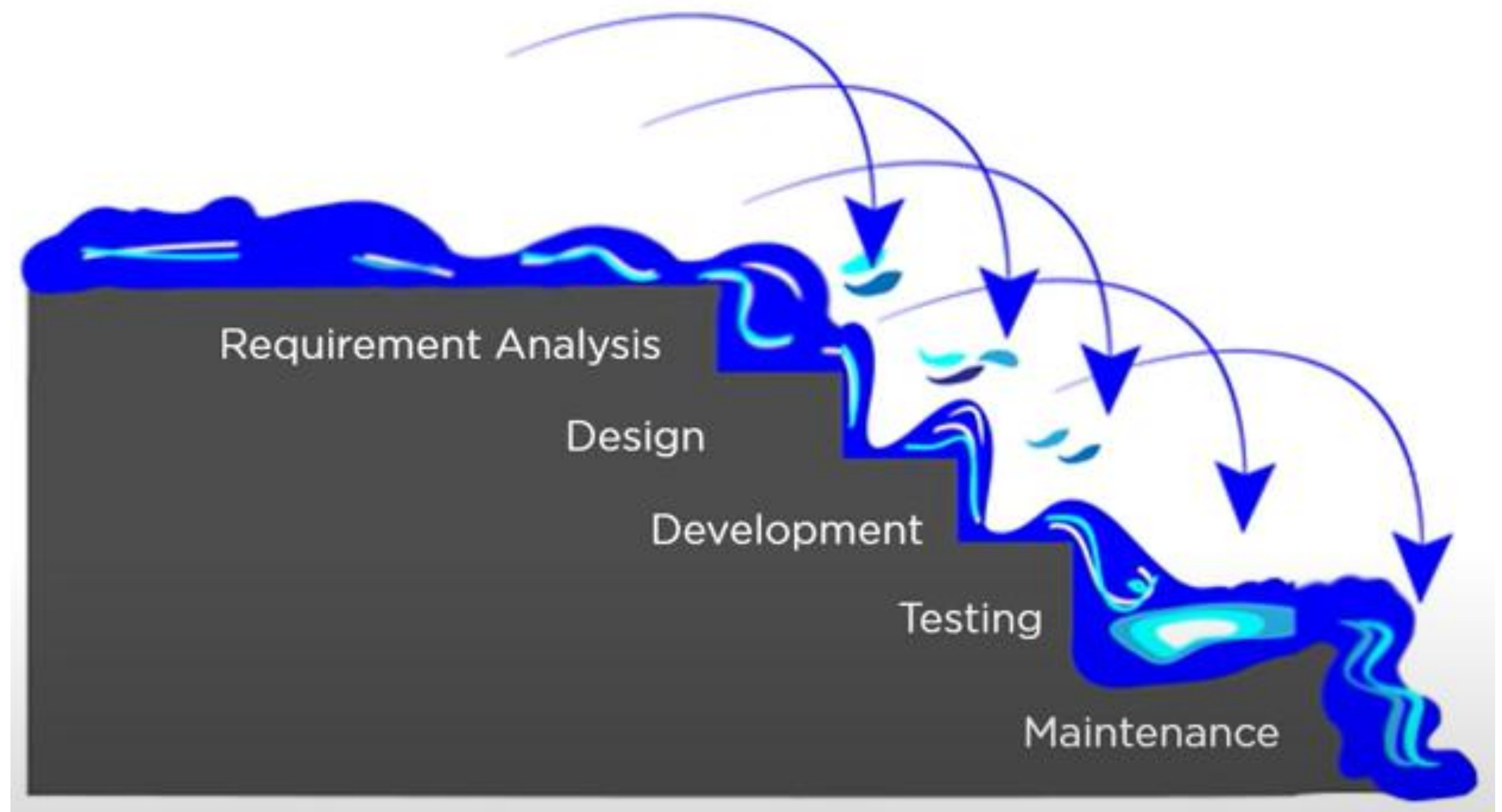


Software Delivery Process

PROJECT TEAM



WATERFALL



**WORK IN
COLLABORATION**



Development
Team



Operations
Team

AGILE OVERVIEW



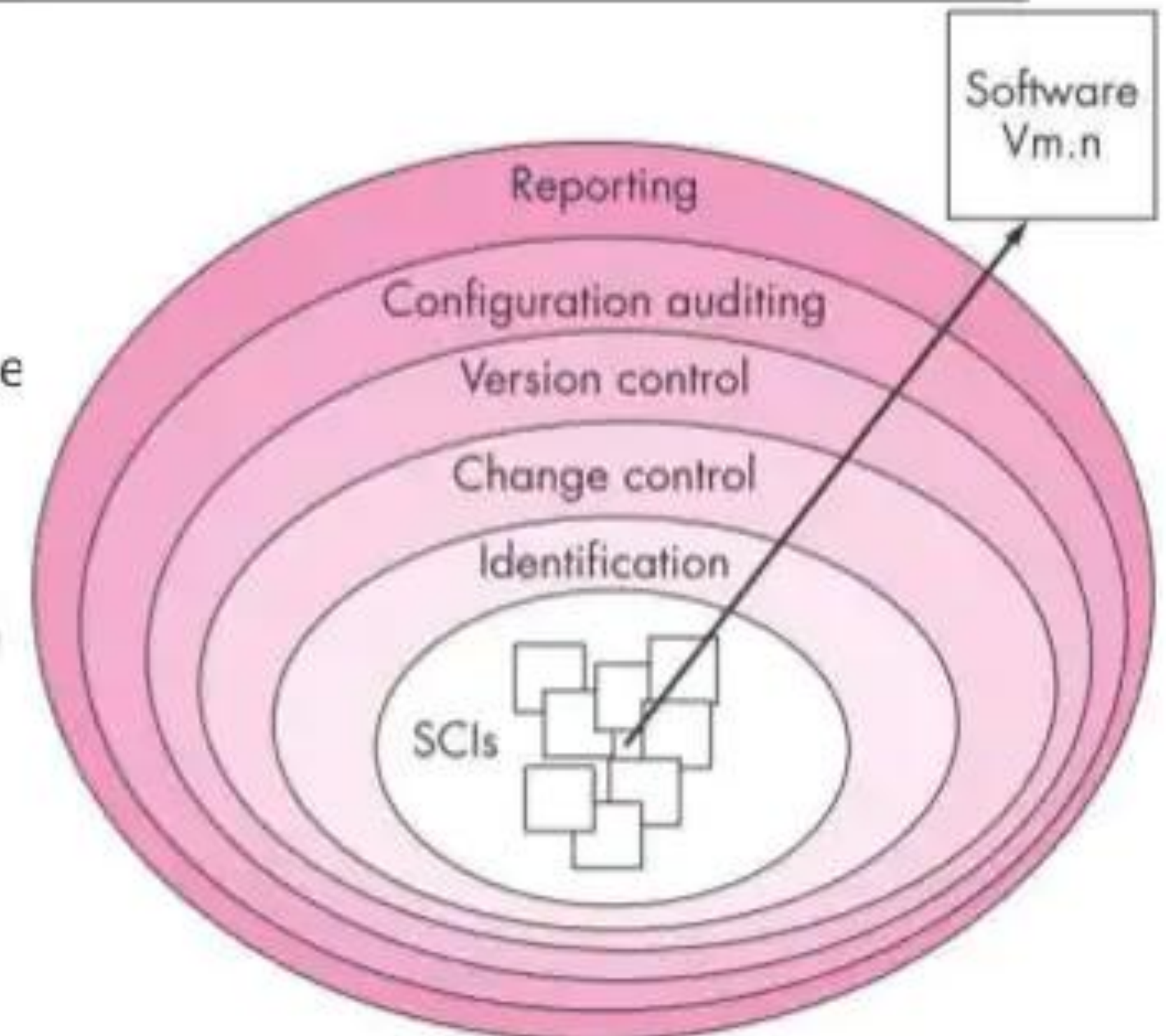
- Agile software development refers to a group of software development methodologies based on iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.
- Agile methods attempt to maximize the delivery of value to the customer and minimize the risk of building products that do not – or no longer – meet market or customer needs.

SOFTWARE CONFIGURATION MANAGEMENT

SCM Process

Primary Objectives:

1. To identify all items that collectively define the software configuration
2. To manage changes to one or more of these items
3. To facilitate the construction of different versions of an application
4. To ensure that software quality is maintained as the configuration evolves over time



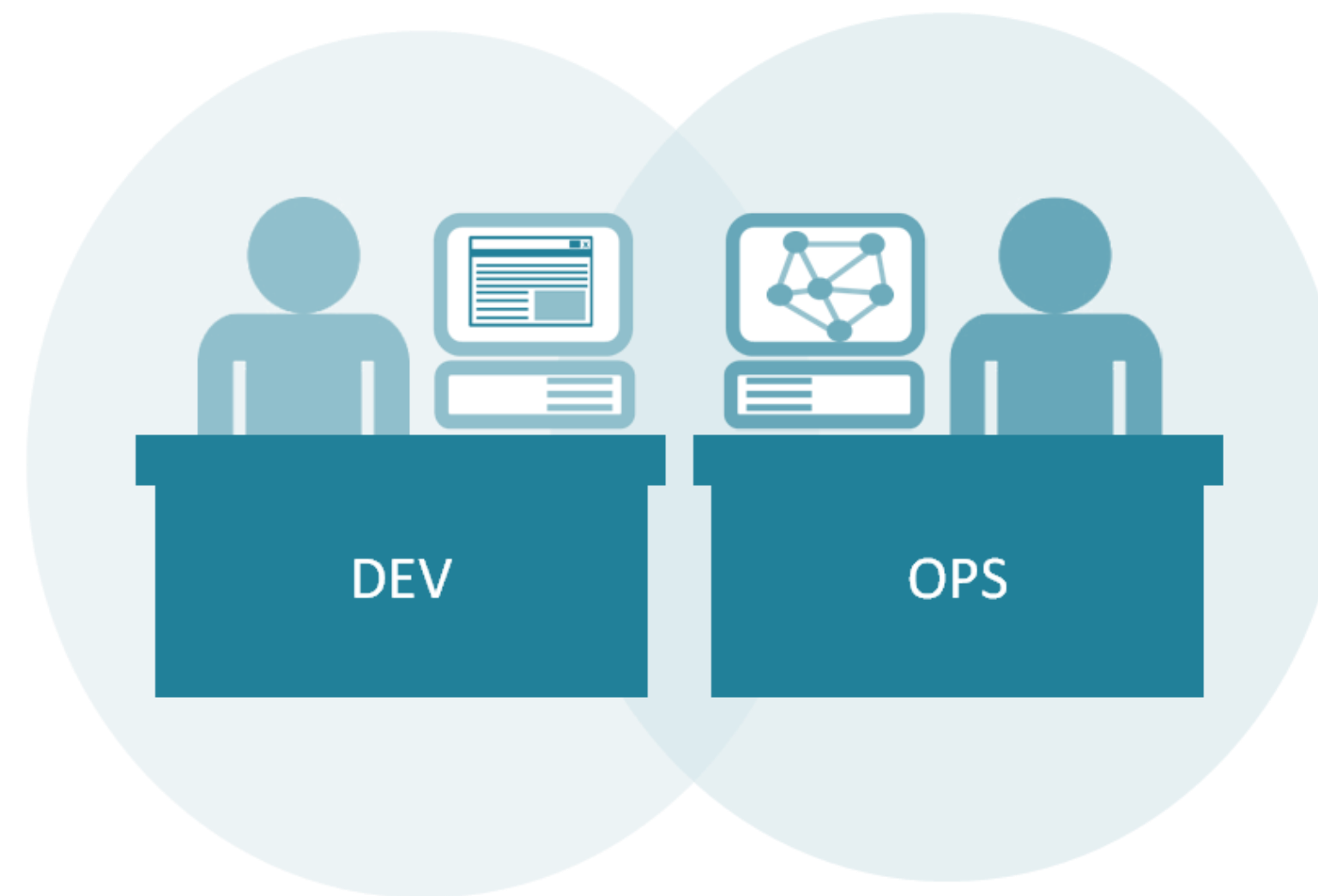
WHAT IS DEVOPS?

- DevOps is a process not a tool.
- DevOps requires tight cooperation between Dev and Ops teams
- Different IT companies define DevOps in a different way:
 1. Developer-friendly operations. IT activities are organized on a way which is more developer-friendly, ie. they are enabling developers to provision infrastructure themselves.
 2. DevOps as one consolidated team of Developers and Operations staff. In this model, Developers take responsibilities of Ops staff and vice-versa.

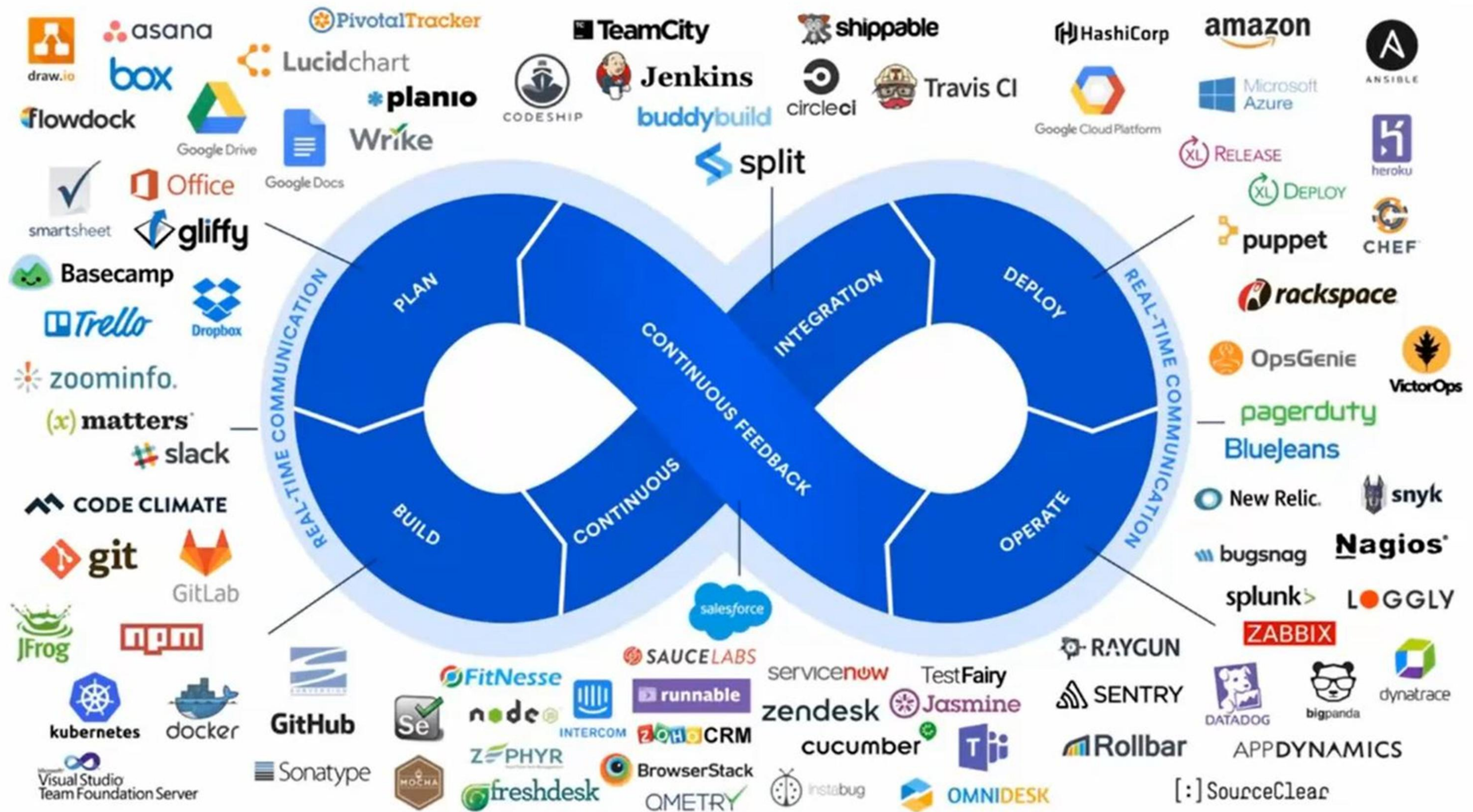


WHAT IS DEVOPS?

- DevOps is all about bringing the latest software changes to end-users fast and frequently with high quality in mind.
- The speed it brings boosts users' satisfaction, beats competition and helps you save a great amount of time and money on fixing problems when they occur.
- DevOps enables delivering software releases in an automated and continuous manner with a click of a button.



**WHAT IS
DEVOPS?**

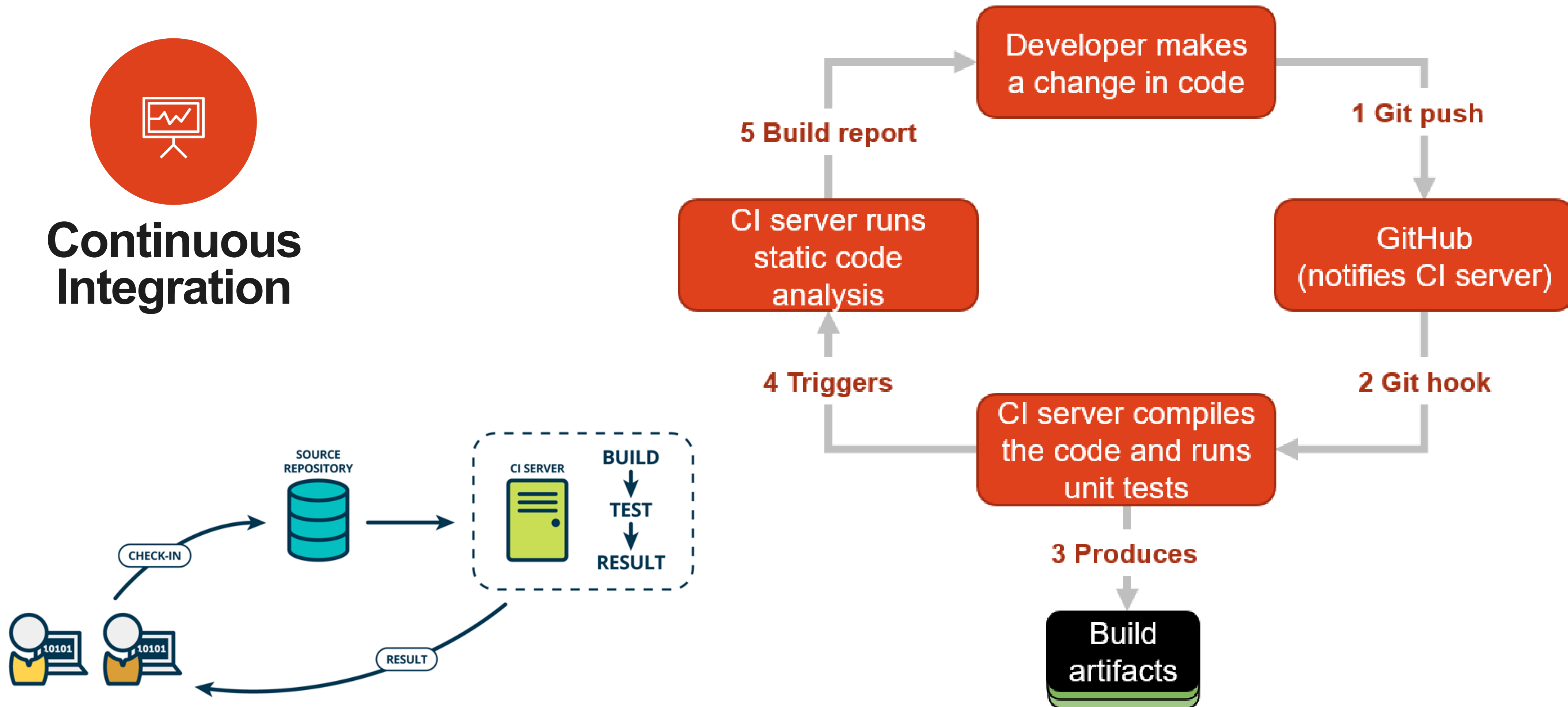


2

Introduction to CI/CD



Continuous Integration



CONTINUOUS INTEGRATION REFERS TO THE BUILD AND UNIT TESTING STAGES OF THE SOFTWARE RELEASE PROCESS. EVERY REVISION THAT IS COMMITTED TRIGGERS AN AUTOMATED BUILD AND TEST.



Continuous Integration

- **CI benefits:**
 - Improve Developer Productivity
 - Find and Address Bugs Earlier
- **How to implement CI?**
 - Keep everything under version control
 - Create automated builds
 - Get agreement from your team for the new way of working

- **Some tools:**

- ✓ Jenkins

- ✓ Travis CI

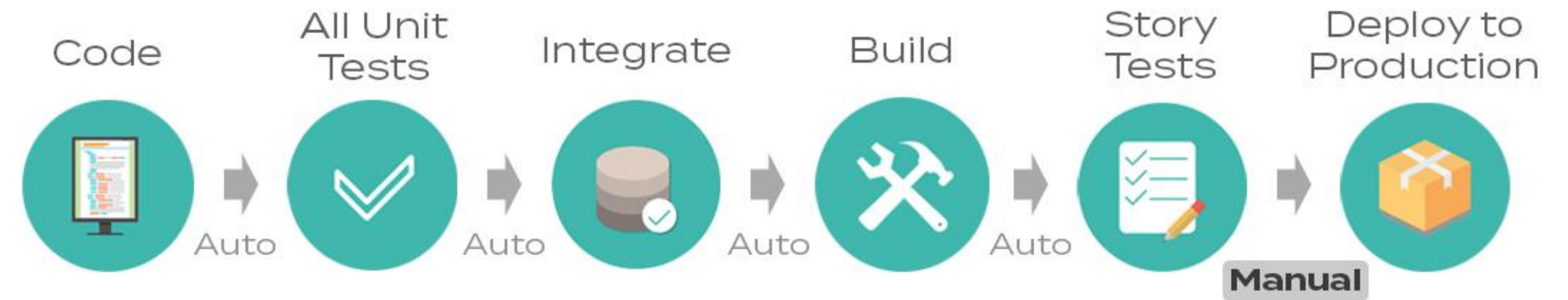
- ✓ Bamboo,

- ✓ Team City





Continuous Delivery

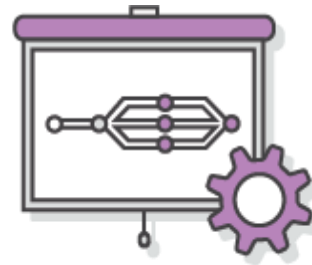


- With Continuous Delivery code changes are automatically built, tested, and prepared for a release to production
- It expands upon Continuous Integration by deploying all code changes to a staging environment after build and tests are completed with success
- When implemented properly, developers will always have a deployment-ready build artifact that has passed through a standardized test process



Continuous Delivery

- **Automate the Software Release Process**



Continuous delivery lets your team automatically build, test, and prepare code changes for release to production so that your software delivery is more efficient and rapid.

- **Improve Developer Productivity**



These practices help your team be more productive by freeing developers from manual tasks and encouraging behaviors that help reduce the number of errors and bugs deployed to customers.

- **Find and Address Bugs Earlier**

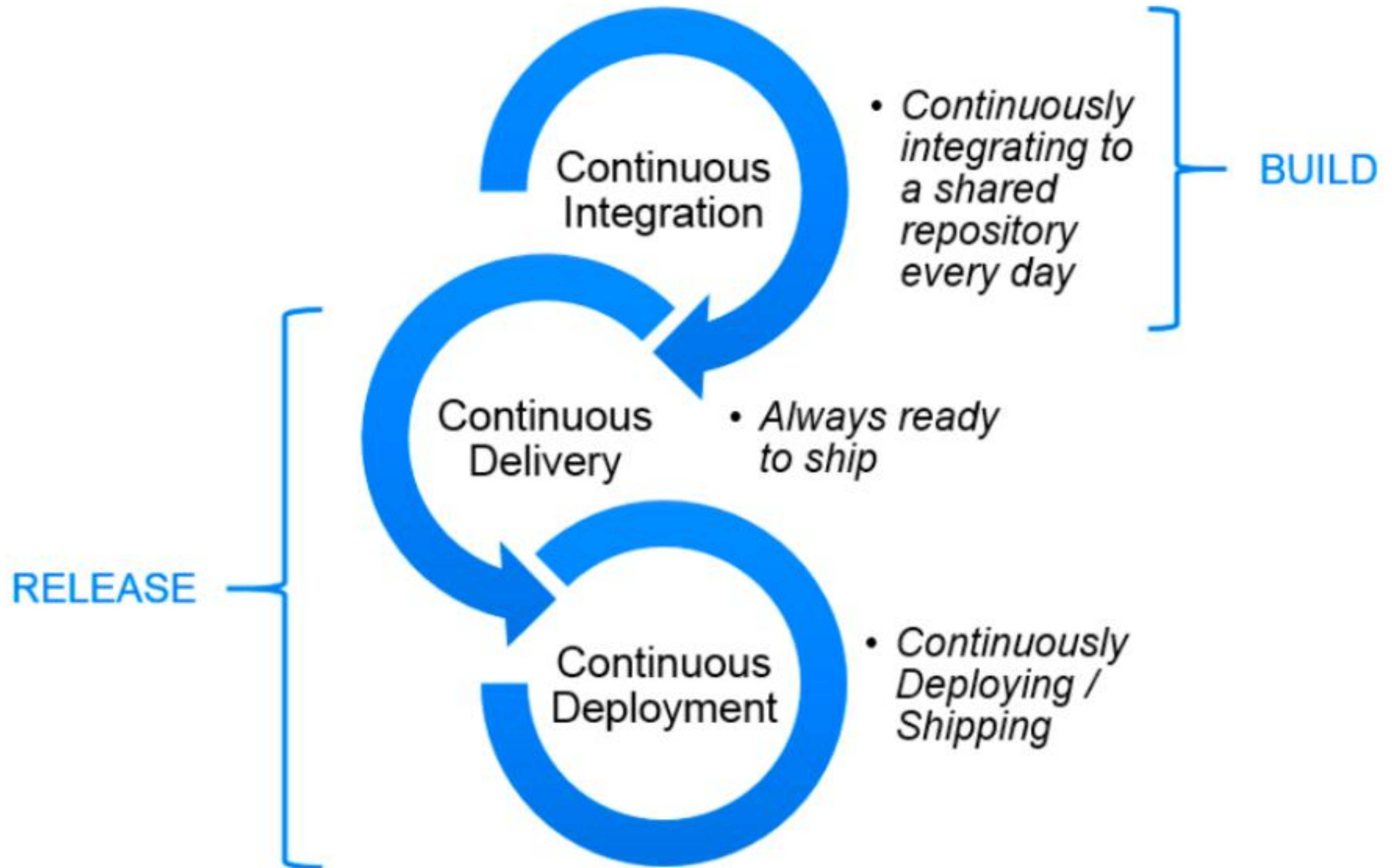


Your team can discover and address bugs earlier before they grow into larger problems later with more frequent and comprehensive testing. Continuous delivery lets you more easily perform additional types of tests on your code because the entire process has been automated.

- **Deliver Updates Faster**



Continuous delivery helps your team deliver updates to customers faster and more frequently. When continuous delivery is implemented properly, you will always have a deployment-ready build artifact that has passed through a standardized test process.



CONTINUOUS MONITORING & LOGGING

- Monitoring & Logging starts in development! The same tools that monitor the production environment can be employed in development to spot problems before they hit production.
- Two kinds of monitoring are required for DevOps: server monitoring and application performance monitoring.
- Some tools for the job: Nagios, Splunk, Datadog

splunk>

Nagios®

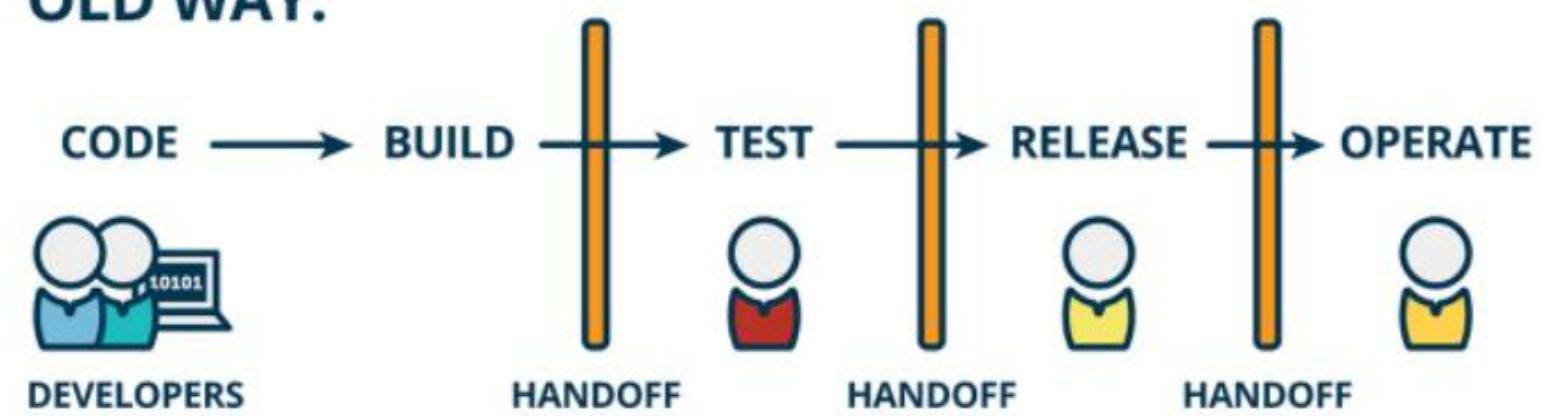


pagerduty
Digital Operations Management

Traditional

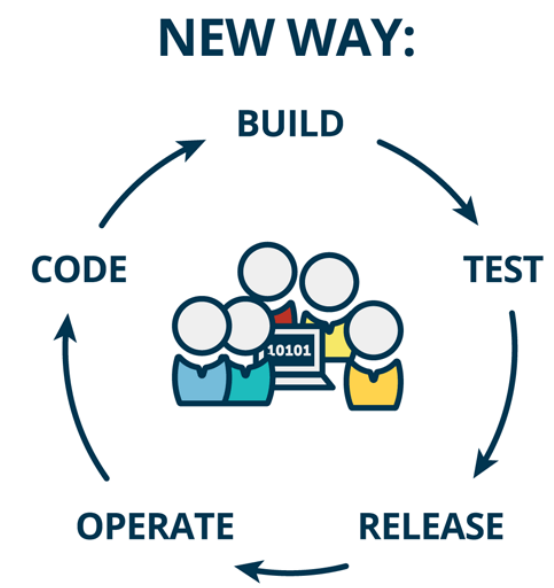
- ⦿ Waterfall approach
- ⦿ Different teams for requirements, development, test and operations
- ⦿ Clear separation between business and IT (demand/supply)
- ⦿ Release 2-3 times a year
- ⦿ Budget/cost drive

OLD WAY:



SOFTWARE DELIVERY

Mode



- ⦿ Agile, scrum
- ⦿ Multidisciplinary teams where all disciplines work together on small piece to deliver
- ⦿ Business, development and operations in one team
- ⦿ Release multiple times a day
- ⦿ Value stream driven

"Our highest priority is to satisfy the customer through early and **continuous delivery** of valuable software"

1st principle behind agile manifesto



Core principles

1. Application and End-User focus
2. Collaboration
3. Performance orientation
4. Development speed
5. Service orientation
6. Automation
7. Monitor everything

- Abundance of tools and technologies



AUTOMATE EVERYTHING



- Define relevant metrics & Set short-term milestones



- And Spread the Word



DEVOPS ADVANTAGES



Time taken to create and deliver software is reduced



Complexity of maintaining an application is reduced

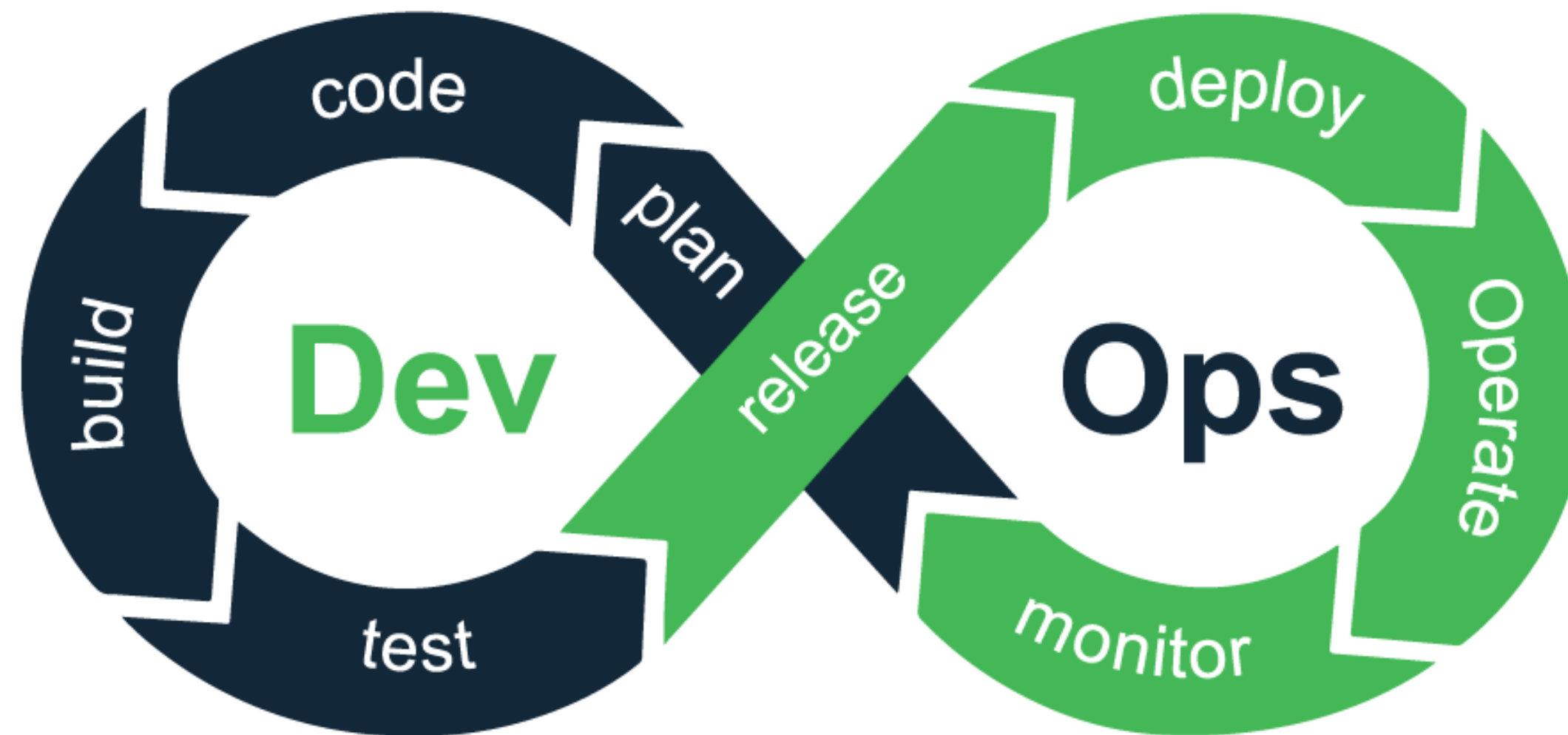


Improved collaboration between developers and operations team

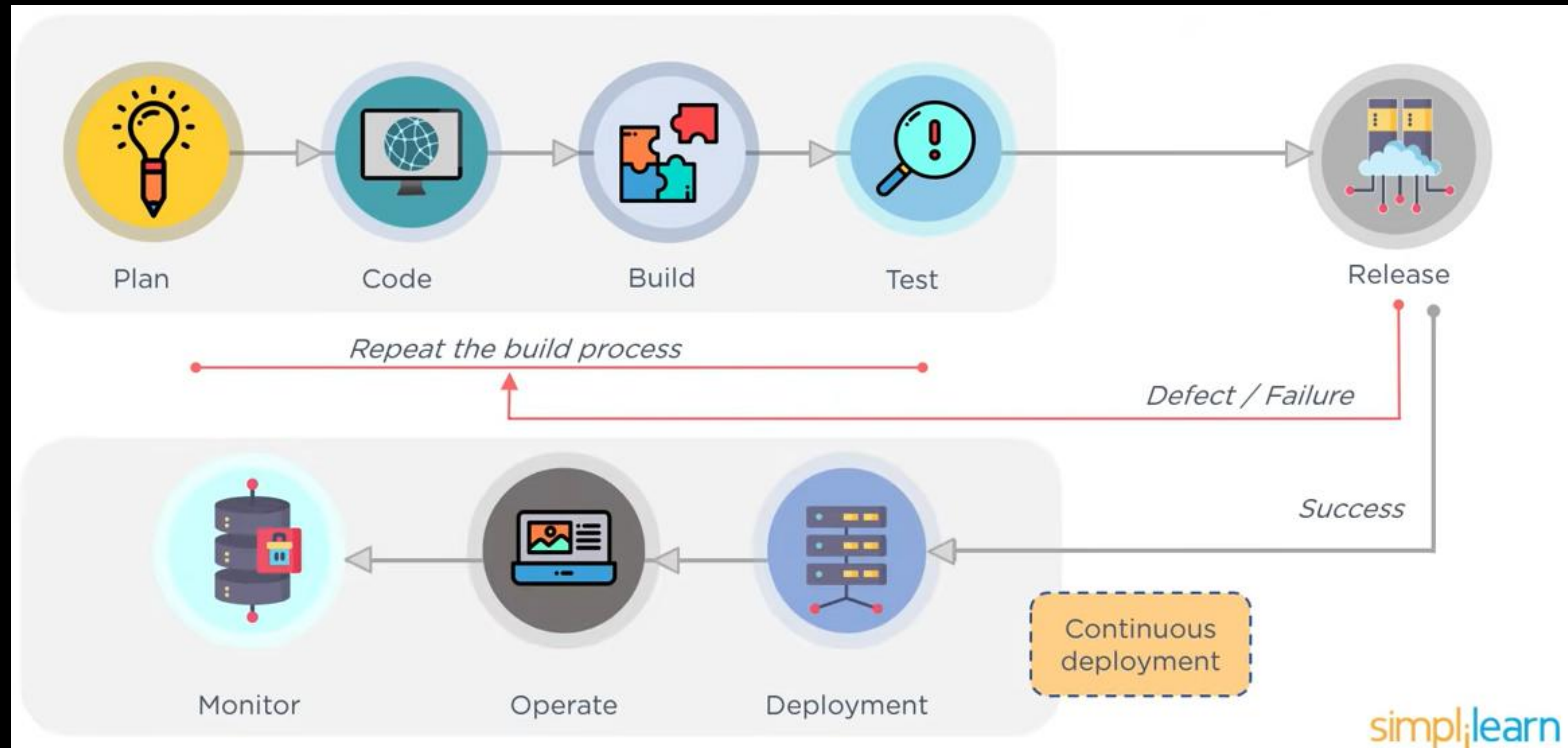


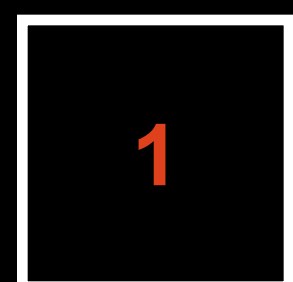
Continuous integration and delivery ensure faster time to market

SO, WHAT IS
DEVOPS



DevOps Flow





Life cycle management

WHAT IS LIFE
CYCLE
MANAGEMENT
?



DISCOVERY



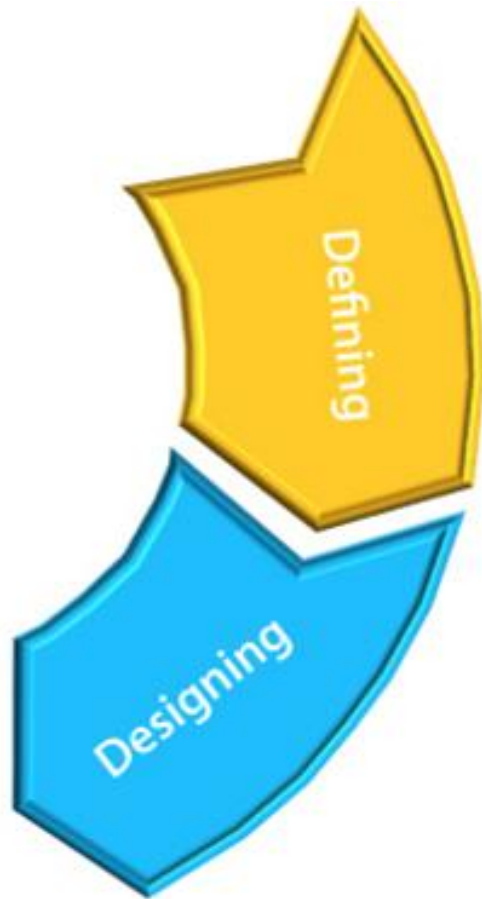
Requirement Analysis



2

Configuration Management

CM - JIRA



Jira

Your work

Projects

Filters

Dashboards

People

Plans

Apps

Create

Search

?

Teams in Space

Classic software project

Scrum: Teams in S...

Board

Roadmap

Backlog

Active sprints

Reports

Issues

Components

Releases

Project pages

Add item

Project settings

Board

Quick Filters

TO DO 5

Engage Jupiter Express for outer solar system travel

SPACE TRAVEL PARTNERS

5

TIS-25

Create 90 day plans for all departments in the Mars Office

LOCAL MARS OFFICE

9

TIS-12

Engage Saturn's Rings Resort as a preferred provider

SPACE TRAVEL PARTNERS

3

TIS-17

Enable Speedy SpaceCraft as the preferred

IN PROGRESS 5

Requesting available flights is now taking > 5 seconds

SEESPACEEZ PLUS

3

TIS-8

Engage Saturn Shuttle Lines for group tours

SPACE TRAVEL PARTNERS

4

TIS-15

Establish a catering vendor to provide meal service

LOCAL MARS OFFICE

4

TIS-15

Engage Saturn Shuttle Lines for group tours

CODE REVIEW 2

Register with the Mars Ministry of Revenue

LOCAL MARS OFFICE

3

TIS-11

Draft network plan for Mars Office

LOCAL MARS OFFICE

3

TIS-15

DONE 8

Homepage footer uses an inline style - should use a class

LARGE TEAM SUPPORT

TIS-68

Engage JetShuttle SpaceWays for travel

SPACE TRAVEL PARTNERS

5

TIS-23

Engage Saturn Shuttle Lines for group tours

SPACE TRAVEL PARTNERS

TIS-15

Establish a catering vendor to provide meal service

LOCAL MARS OFFICE

Release

[Pages](#) / [Demo Project Home](#)

/ Product requirements

User story: Dashboard

Created by Atlassian OnDemand [Administrator] on Dec 19, 2016

Epic link	 SOF-62 - Dashboard To do
Status	DRAFT
Developer	@Arjan
Designer	@Sami
Product manager	@Des C

Primary user story

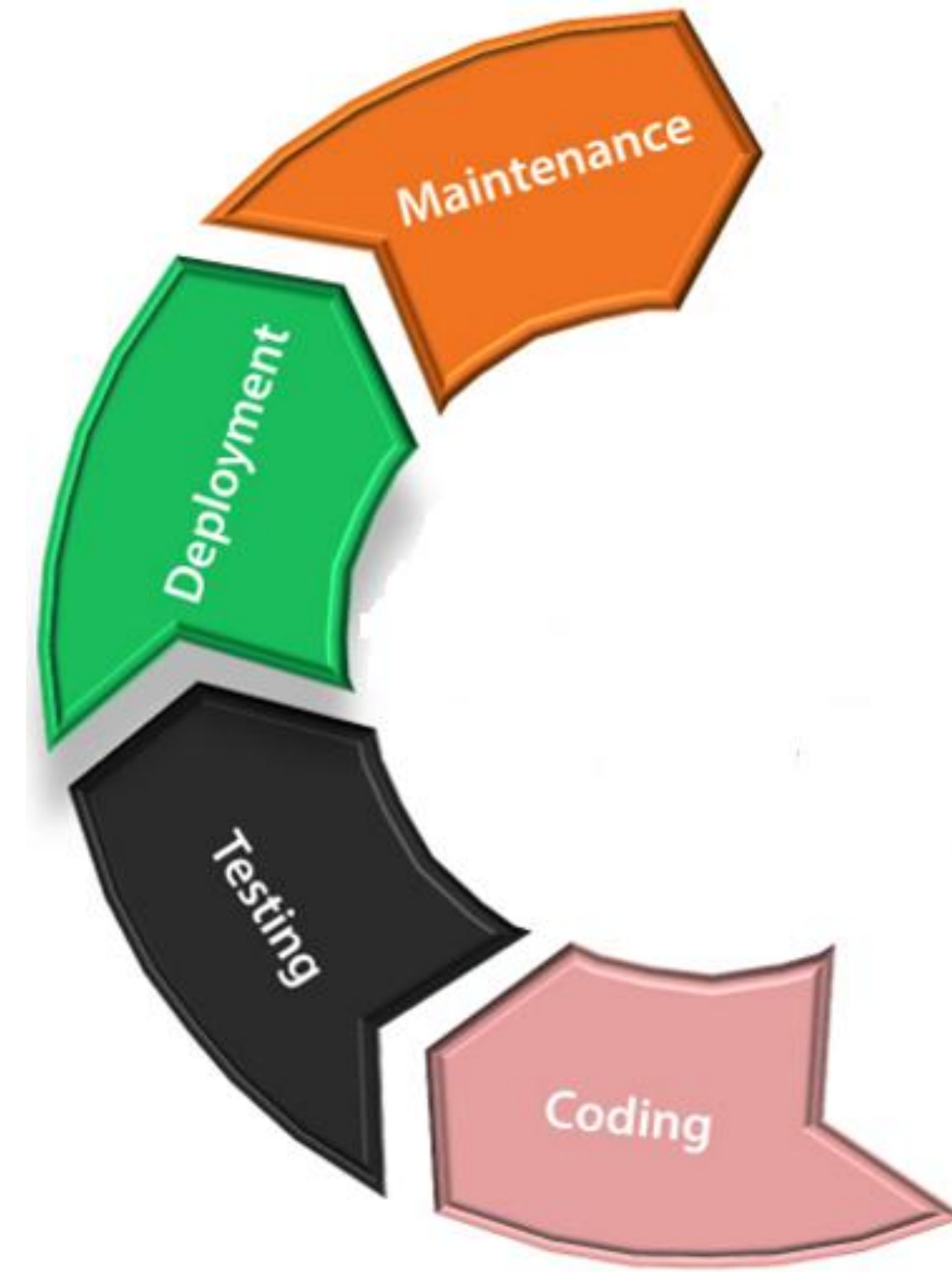
As a user, I want to be see a status of everything related to my work in Confluence, **so that** I can triage my work while on the go.

Requirements

#	Title	User Story	JIRA Issues	Notes	Importance
1	Activity feed	As a user, I want to see the most recent activity related to my work so that I can efficiently	 SOF-34 - Activity feed DONE PRESS	Order pages chronologically with the most recent updates first.	MUST HAVE

3

Source control management



SCM

- Some teams may still not use it!



new



newest



proj_20171030



proj_20171030_works_great



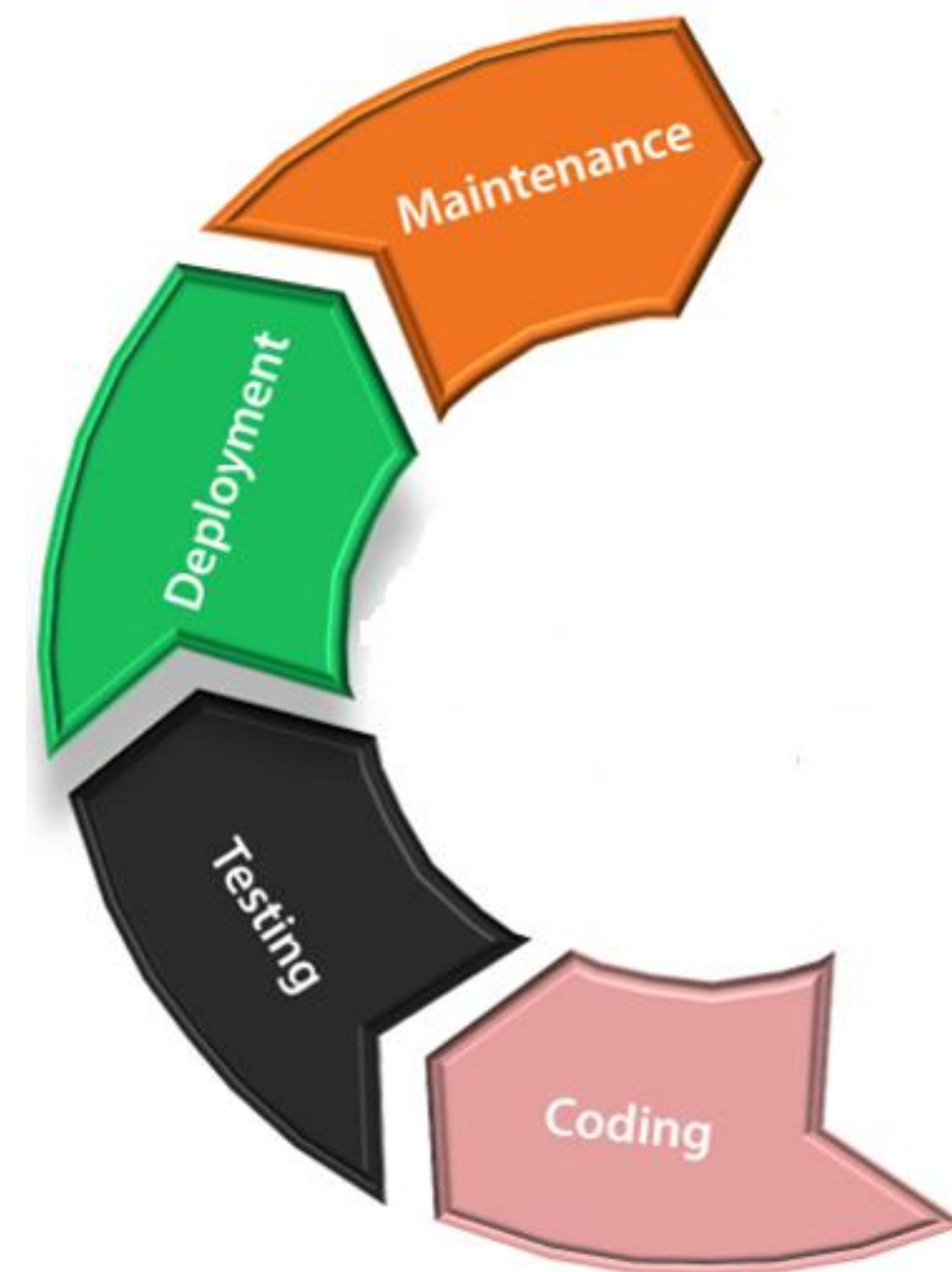
proj_the_best_version

- It needs to store everything is needed to reproduce application environments: application code, third party libraries, documentation, build scripts, tests, database schemas.



4

CI/CD Tools

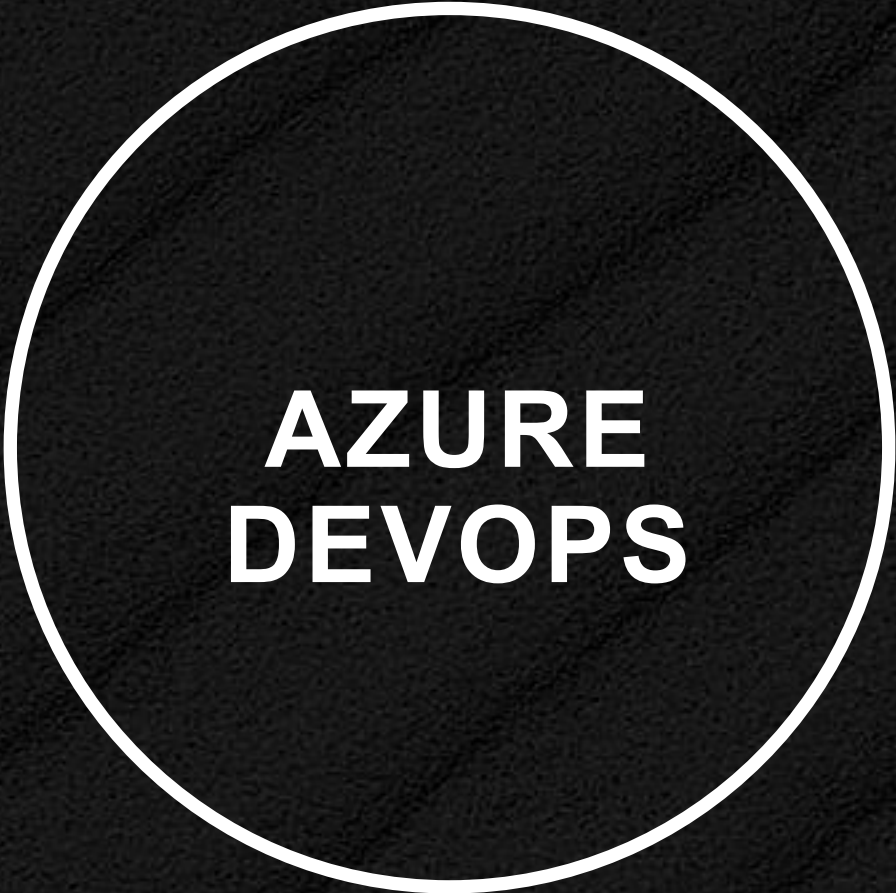


PIPELINE TOOLS

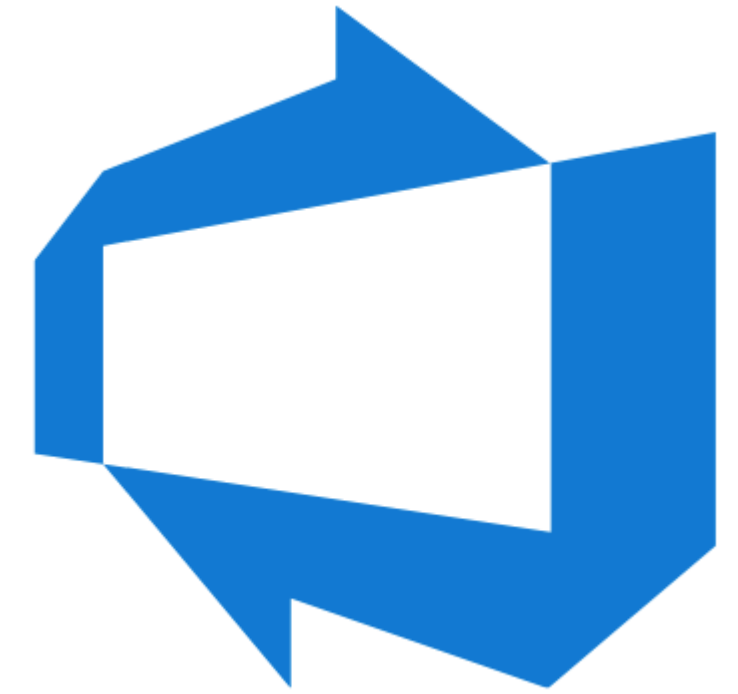


Travis CI





AZURE DEVOPS



- Azure DevOps provides developer services for support teams to plan work, collaborate on code development, and build and deploy applications.
- Azure DevOps supports a culture and set of processes that bring developers and project managers and contributors together to complete software development, code development and build and deploy applications. Azure DevOps supports a culture and set of processes that bring developers and project managers and contributors together to complete software development.

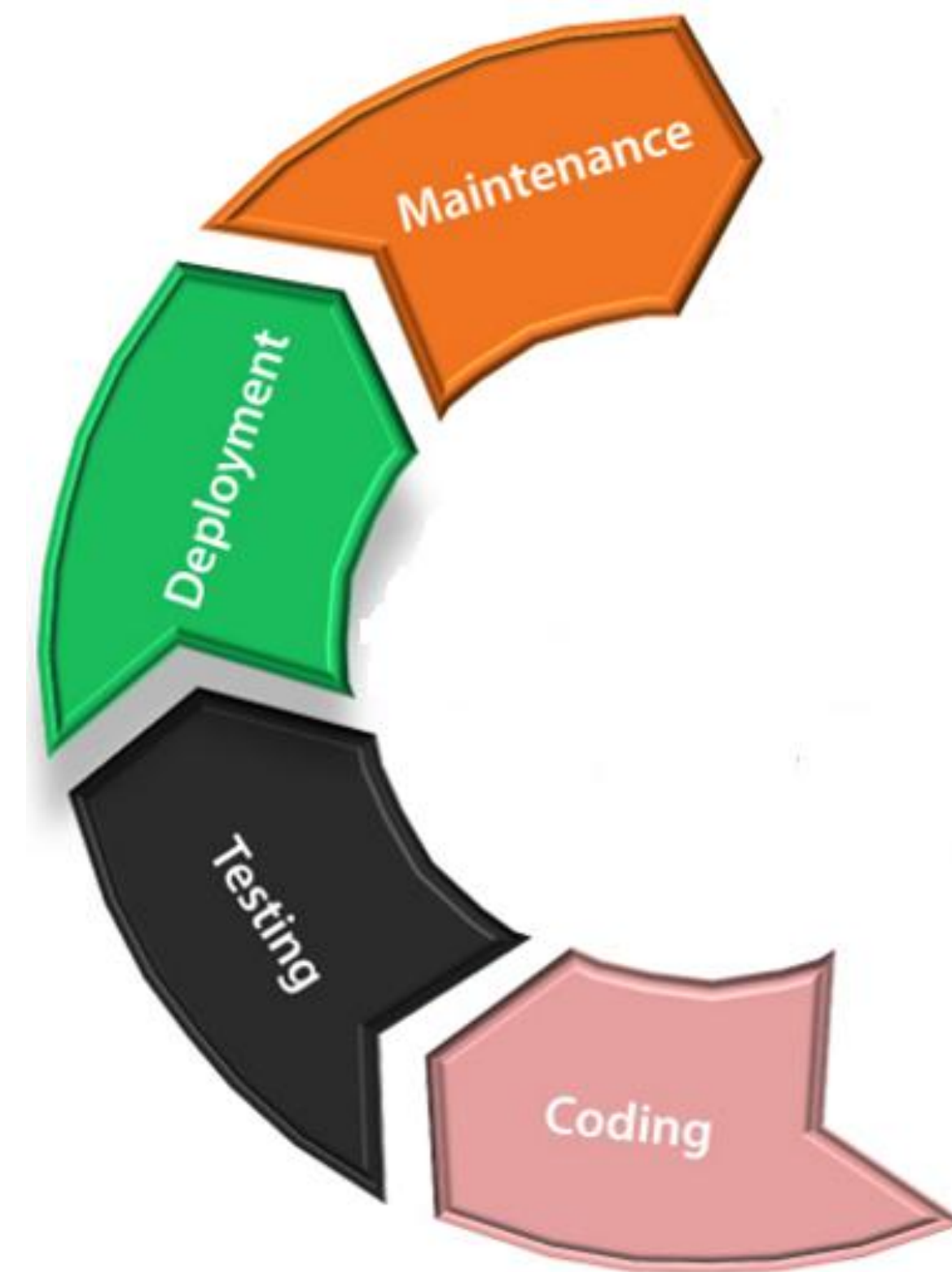


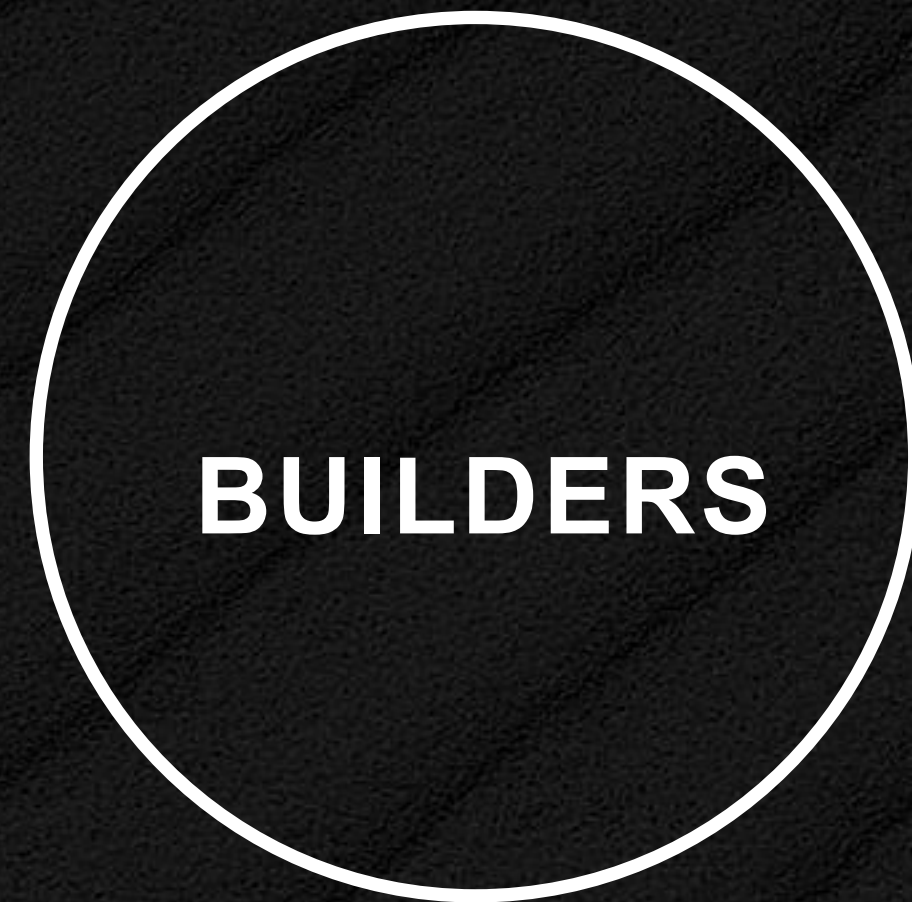
GitHub is a website and cloud-based service that helps developers store and manage their code, as well as track and control changes to their code. To understand exactly what GitHub is, you need to know two connected principles:

- Version Control
- GIT
- CI/CD

5

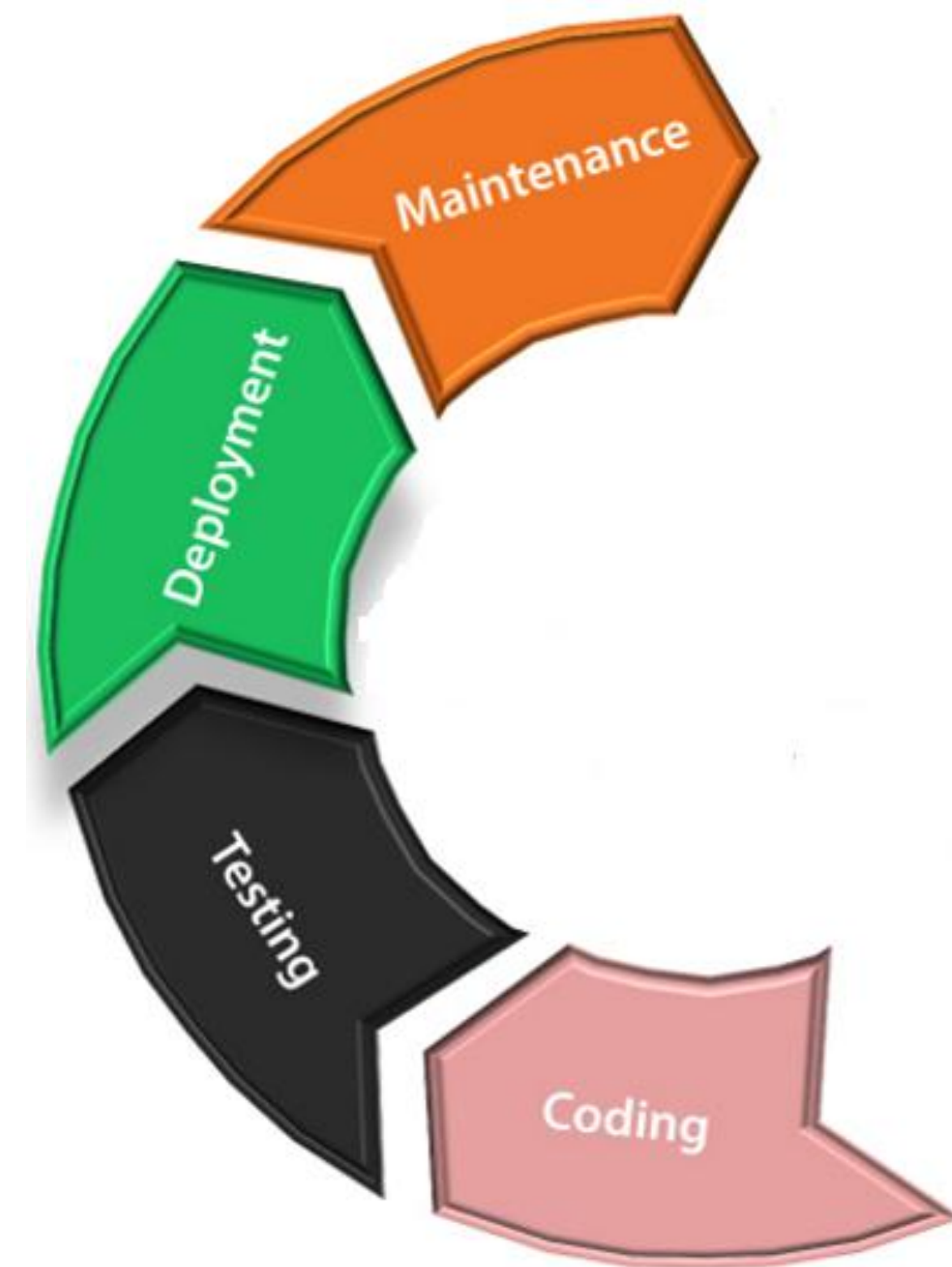
Build tools

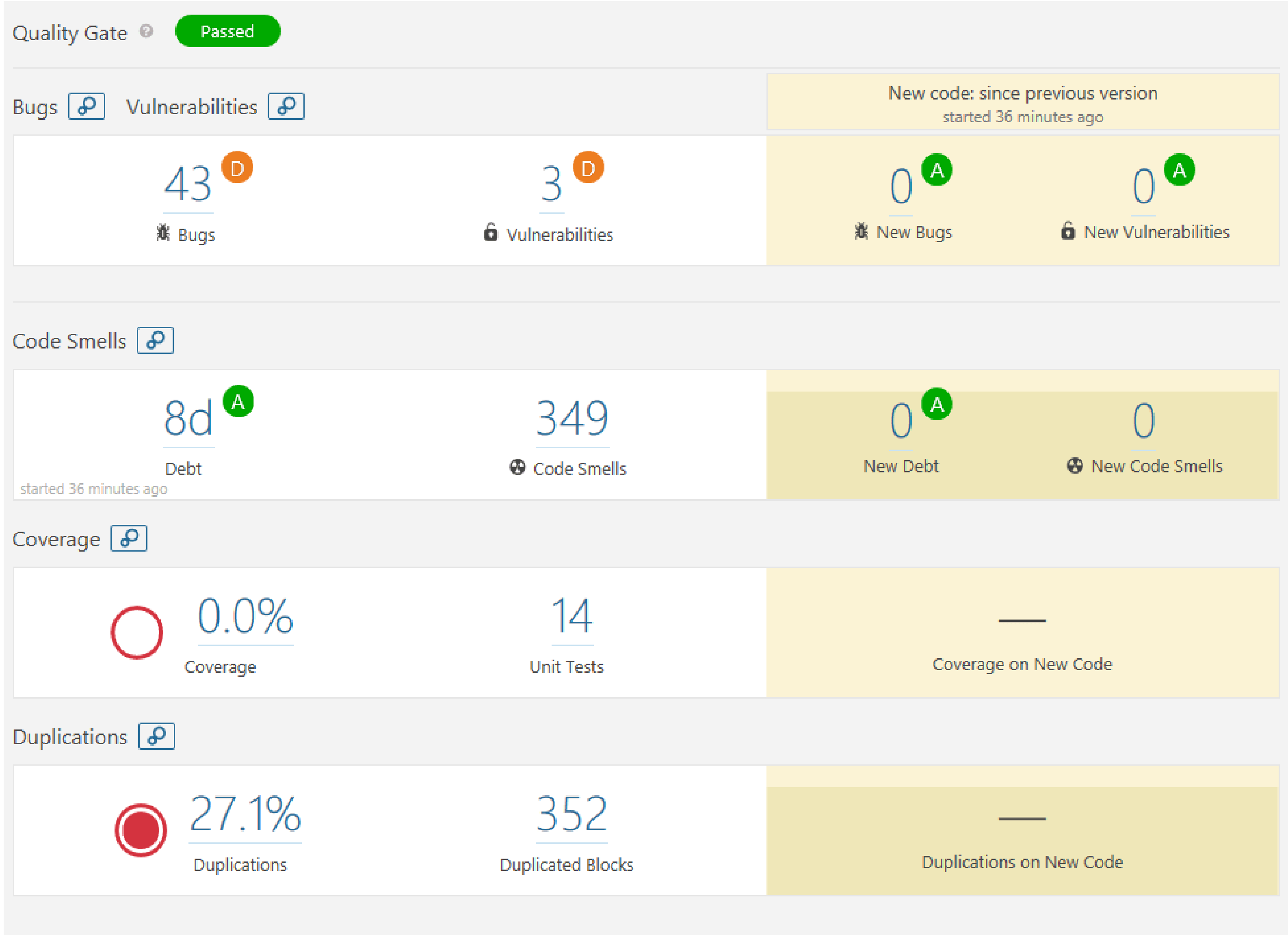




6

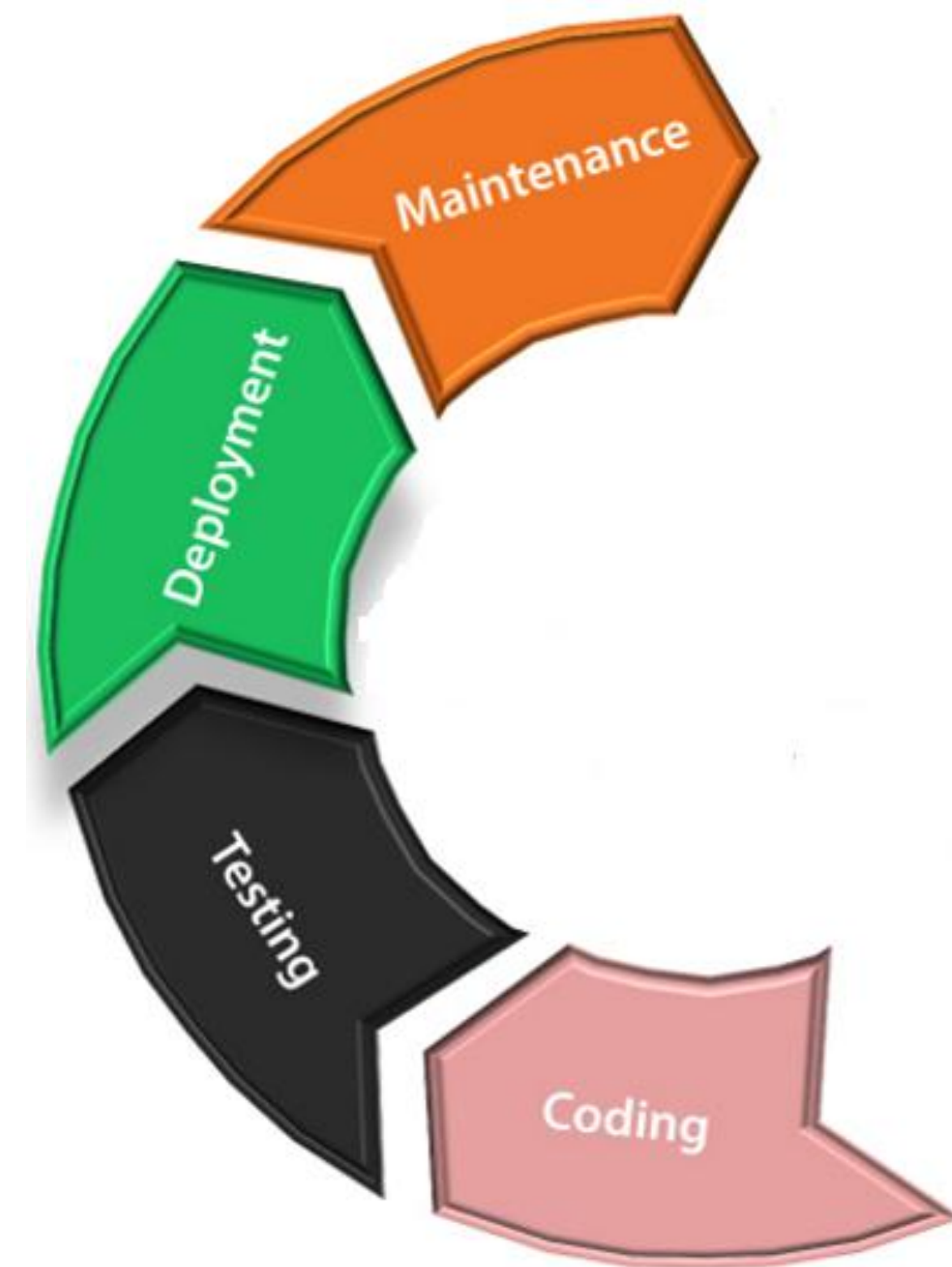
Static code Analyze





7

Artifact repository



ARTIFACT REPOSITORY



JFrog Artifactory

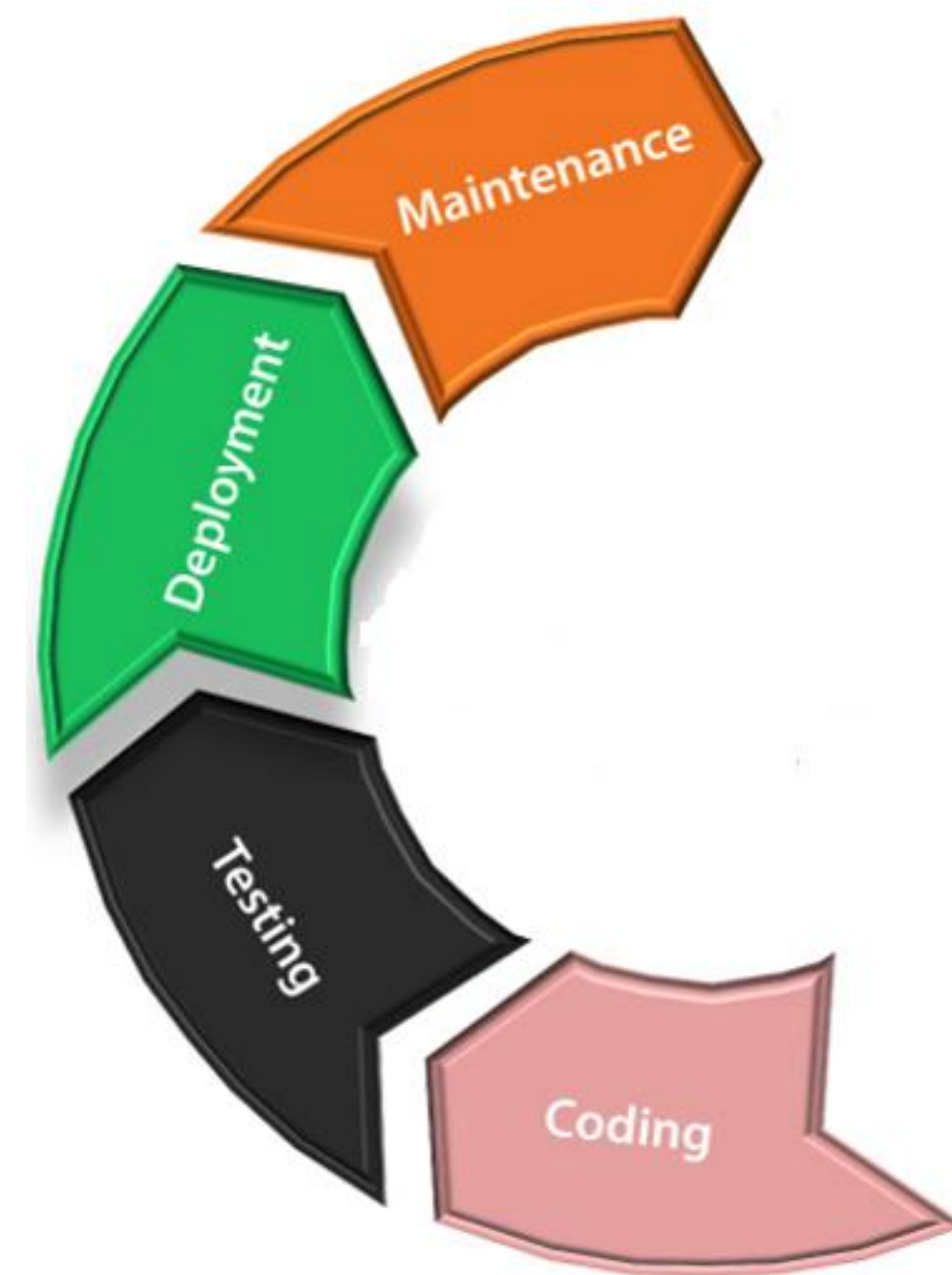
- Artifact are large binary packages that are created throughout the development and release process.
- An artifact repository is a software application designed to manage these artifacts. Using an artifact repository provides consistency to your Continuous Integration/Continuous Development (CI/CD) workflow. It saves teams time and increases build performance.



nexus

8

Deployment

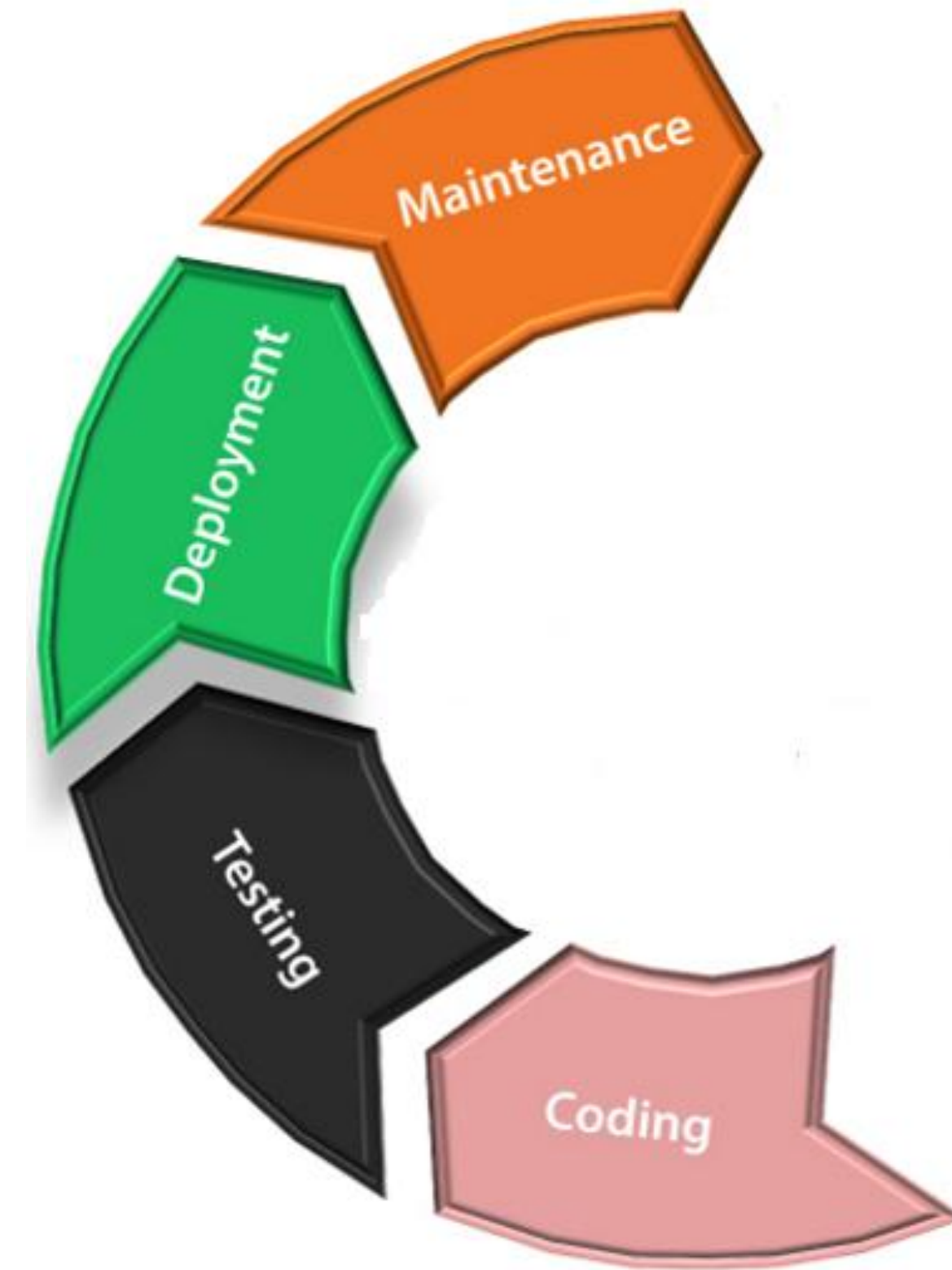


DEPLOYMENT



9

Testing tools



TESTING
TOOLS

cucumber



APACHE
JMeterTM



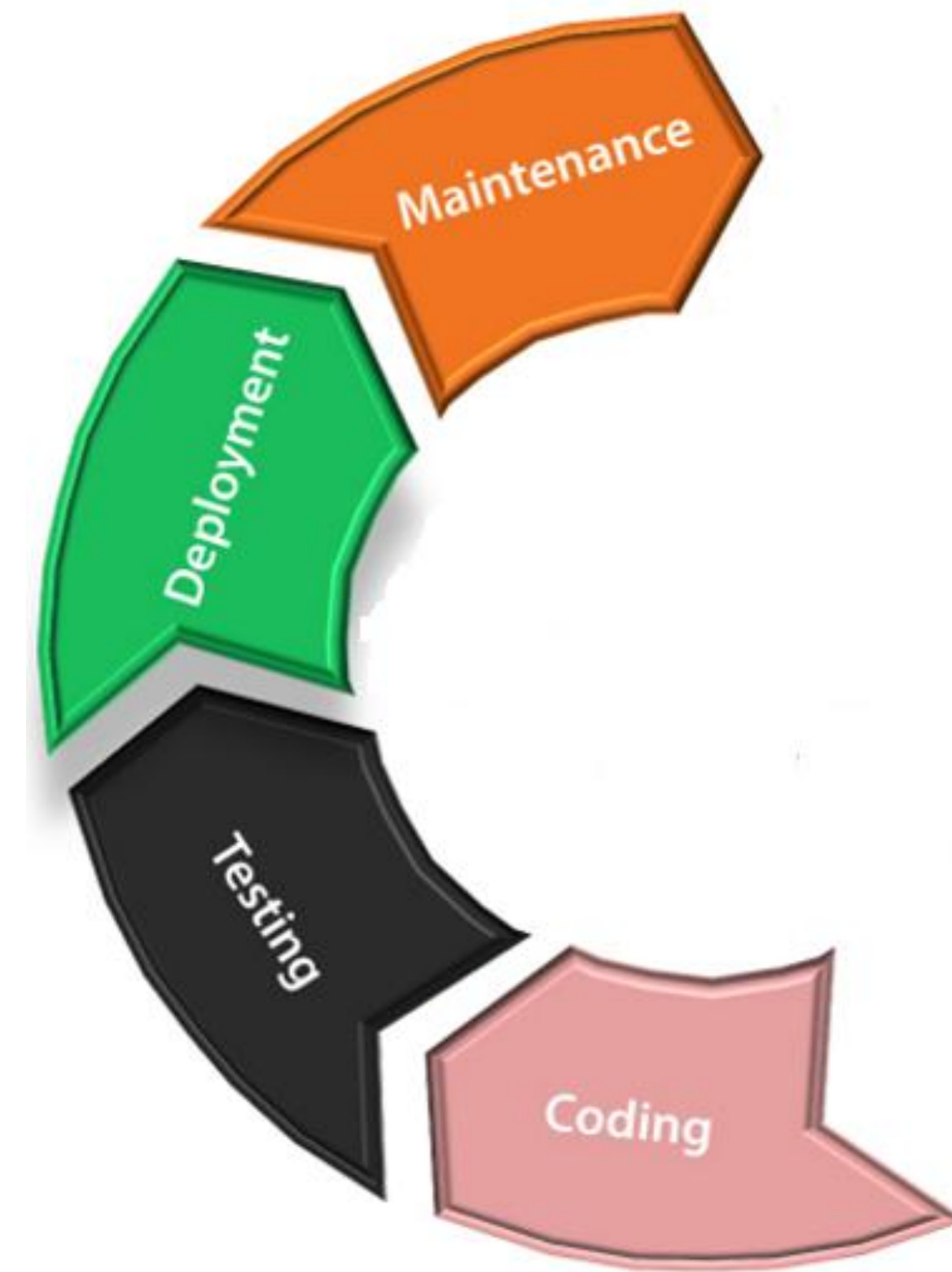
Analysis

VuGen Controller
LoadRunner

JUnit

10

Logging and Monitoring



LOGGING &
MONITORING

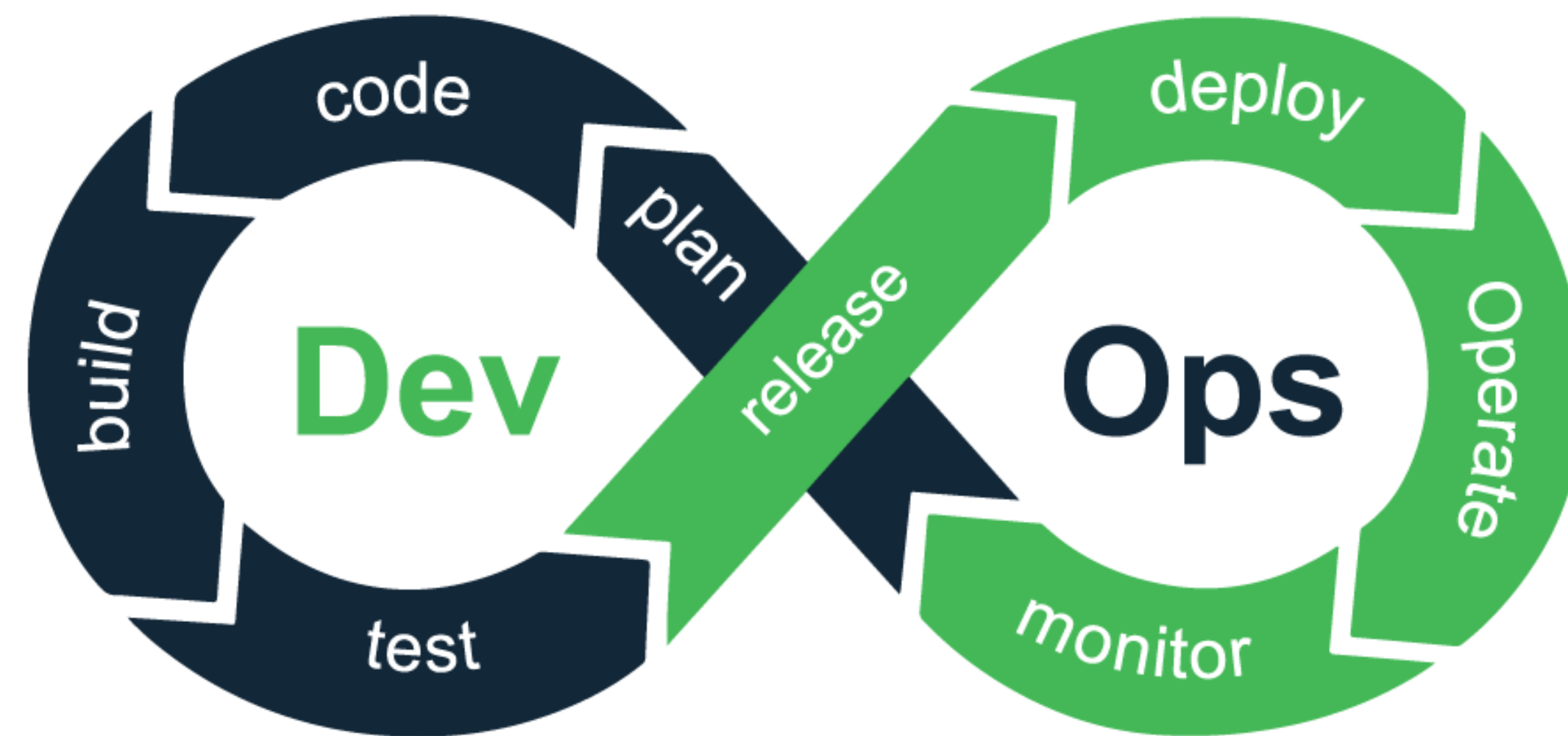
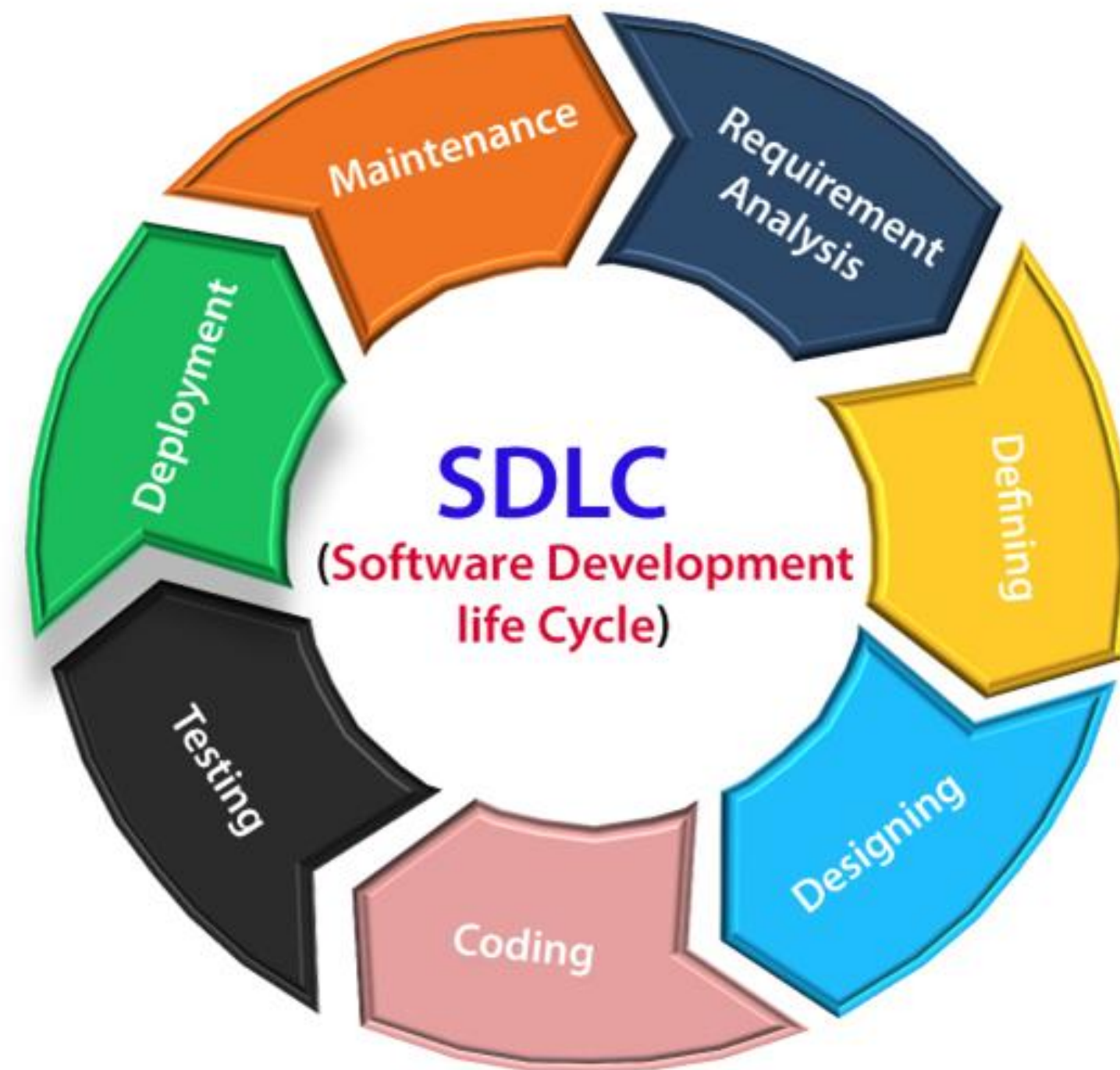


Nagios®



splunk>







Endava DevOps Team